

Pandas Roadmap

Complete Learning Guide for GenAI Developers

Duration: 4 Weeks | **Level:** Total Beginner → Intermediate

Total Hours: 60-80 hours | **Goal:** Master Pandas for GenAI Development

Why Pandas for GenAI Developers?

- [+] Data Preprocessing: Clean and prepare datasets before training LLMs
- [+] EDA (Exploratory Data Analysis): Understand your data before feeding to models
- [+] Feature Engineering: Transform raw data into ML-ready features
- [+] Dataset Management: Handle CSV, JSON, and Parquet files efficiently
- [+] Interview Ready: 90% of DS/ML interviews include pandas questions

WEEK 1: Pandas Fundamentals (Beginner)

Focus: Understanding core concepts | **Time:** 15-20 hours

- Installation & Setup
- Series Basics
- DataFrame Creation
- Data Types in Pandas
- Reading Data Files (CSV, JSON)
- DataFrame Inspection (head, tail, info, describe)
- Column Selection & Access
- Basic Indexing

Mini Project: Dataset Explorer

Load any CSV dataset, explore structure, display info, access columns, export findings

WEEK 2: Data Cleaning & Preprocessing

Focus: Make messy data clean | **Time:** 15-20 hours

- Handling Missing Data (NaN, dropna, fillna)
- Removing Duplicates
- Data Type Conversion
- String Operations & Text Cleaning
- Datetime Parsing & Handling
- Outlier Detection Basics
- Data Normalization & Scaling
- Reset & Rename Operations

Mini Project: Data Cleaning Pipeline

Load messy dataset, apply cleaning, document before/after, export cleaned data

WEEK 3: Data Analysis & Transformation

Focus: Extract insights and reshape data | **Time:** 15-20 hours

- Indexing Deep Dive (loc, iloc, at, iat)
- Boolean Indexing & Filtering
- GroupBy Operations
- Merging DataFrames (merge, join, concat)
- Reshaping Data (pivot, melt, stack, unstack)
- Aggregations (sum, mean, count, std)
- Sorting & Ranking
- Window Functions

Mini Project: Sales Analysis Dashboard

Filter, group, pivot sales data; merge datasets; generate insights

WEEK 4: GenAI Data Prep & Advanced Topics

Focus: Optimize for ML/GenAI workflows | **Time:** 15-20 hours

- Reading Large Datasets (chunks, dtypes optimization)
- Categorical Data Handling
- Time Series Basics
- Apply Functions (map, apply, applymap)
- Performance Optimization Tips
- Export Data (CSV, JSON, Parquet, Excel)
- Data Pipeline Creation
- Real-World GenAI Dataset Prep

Capstone Project: End-to-End GenAI Data Pipeline

Load raw dataset, clean, transform, create train/test split, export to Parquet, build pipeline

Interview Questions by Level

Week 1 Level:

- What's the difference between Series and DataFrame?
- How do you read a CSV file?
- How do you access columns in a DataFrame?

Week 2 Level:

- How do you handle missing values?
- How do you remove duplicates?
- How do you convert data types?

Week 3 Level:

- How do you group data and aggregate?
- What's the difference between merge() and join()?
- How do you reshape data with pivot()?

Week 4 Level:

- How do you handle large datasets efficiently?
- What's categorical data and why use it?
- How would you prepare data for an ML model?

Pro Tips for Success

[OK] Practice with real data - Use Kaggle datasets

[OK] Type along - Don't just read, write code

[OK] Build projects - Apply learning immediately

[OK] Refer to docs - Pandas docs are excellent

[OK] Join communities - Stack Overflow, Reddit r/datascience

[OK] Interview prep - Practice LeetCode pandas problems

Next Steps After Pandas:

1. **NumPy** - Numerical computing (parallel to pandas)
2. **Scikit-learn** - Machine learning (uses pandas)
3. **Data Visualization** - Matplotlib, Seaborn
4. **SQL** - Query databases efficiently
5. **PySpark** - Big data processing

Pandas Roadmap for GenAI Developers | 4 Weeks of Structured Learning

Track your progress in your HTML tracker!